MARY K. BEALS, Ph.D.

Assistant Professor of Practice — Department of Biological Sciences & Chemistry

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PROFESSIONAL SUMMARY

Student-centered biology educator, curriculum architect, instructional designer, and AI-integrated learning practitioner with more than 20 years of university teaching experience across lecture, laboratory, hybrid, and fully online modalities. Current work centers on culturally responsive andragogy, RSI-driven course design, AI-literate pedagogy, data-informed student success, and practice-ready learning experiences in Biology, Botany, Ecology, Anatomy & Physiology, and General Biology. Committed to expanding research literacy, technology access, and workforce-relevant STEMM skills for students in HBCU learning environments.

CURRENT ACADEMIC APPOINTMENTS, FELLOWSHIPS & PROFESSIONAL DESIGNATIONS

Assistant Professor of Practice | *Southern University and A&M College, Department of Biological Sciences & Chemistry | 2025-Present*

- Lead instructor and course designer for biology courses including Botany, Ecology, General Biology, Anatomy & Physiology, and laboratory-based STEM education.
- Develops RSI-aligned, student-centered course structures with transparent rubrics, active learning, digital learning resources, and equity-focused student support.

CDS-Exchange Community Fellow | *Collaborative Data Science Exchange | 2025-2026*

- Engages in cross-institutional data science initiatives focused on computational thinking, AI integration, community-based research, and STEM pathways for HBCU students.

AI SuperUser / Stipendiary Appointment | *SMART Technology Innovations Center, Tennessee State University | 2025-Present*

- Applies high-fidelity prompt design, ethical AI workflows, bias/accuracy evaluation, and AI-supported productivity practices to academic and professional contexts.

NST Fellow | *2026*

- Participates in professional learning and STEM education development focused on teaching innovation, student success, and science education practice.

INSTRUCTIONAL INNOVATION & DIGITAL SCHOLARSHIP

- AI-Integrated Biology Curriculum: develops course activities, study resources, student-facing tools, and AI disclosure frameworks that promote ethical AI use, scientific accuracy, and student accountability.
- NotebookLM-Powered Digital Lab Manuals & Notebooks: centralizes laboratory protocols, PPE/safety guidance, micro-rubrics, reflective prompts, and student-owned artifacts for Botany and Ecology.
- AI Course Companion / Grace Lab / Nightingale Coach GPT Concepts: designs AI-supported learning assistant frameworks for student coaching, concept review, and discipline-specific academic support while preserving human oversight.
- Low-Cost and Accessible Laboratory Design: creates household-materials and remote-ready lab activities in Ecology and General Biology that support inquiry, modeling, data visualization, and experimental design.
- Canvas Blueprints & Accessible Course Templates: builds SU-branded Canvas structures, RSI blocks, module pages, assignment templates, rubric frameworks, and student-facing learning resources.
- Learning Analytics and Student Support: uses lightweight data review, grade patterns, item analysis, feedback cycles, and targeted outreach to inform scaffolding and improve course navigation.

SELECTED COURSE & CURRICULUM LEADERSHIP

- General Education Botany Curriculum Redesign: developing a student-centered, general education Botany course that transforms STEM-major content into accessible, engaging, real-world plant science learning.
- Asynchronous Ecology Lab Manual: authoring and refining a 10-lab Ecology manual incorporating remote/in-person options, biodiversity analysis, biogeochemical cycles, environmental data, and RSI-compliant engagement.
- General Biology Non-Majors Lab Manual: editor/coordinator for active-learning redesign, common assessments, accessible assignment structures, and rubric alignment.

- SBIO 101B/102B RSI & Course Content Redesign Team (QEP): contributor to module alignment, RSI documentation, discussion/feedback standards, and course quality improvement.
- Course-Based Undergraduate Research Experience (CURE) Development: advancing ecology, botany, and environmental biology activities that connect students to data science, AI literacy, and community-relevant STEMM questions.

SELECTED COURSES TAUGHT / REDESIGNED

- Ecology (Online, 2021-Present): fully asynchronous lecture/lab model with RSI cadence, low-cost labs, digital lab manual, practicums, AI-literate tasks, and environmental data applications.
- Botany (Lecture & Lab, 2010-Present; 2025-2026 refresh): modular design with laboratory phases, plant-centered inquiry, micro-rubrics, SU-branded Canvas templates, and digital notebooks.
- General Biology / Majors and Non-Majors: active-learning redesign, common assessments, BioNotes/JagScript-style student engagement tools, and accessible Canvas-based learning resources.
- Anatomy & Physiology I & II: healthcare-aligned learning, safety-first laboratories, skills mapping, standardized rubrics, and student support for pre-health/nursing pathways.
- Laboratory-Based STEM Education: microscopy, biological molecules, cellular processes, genetics, DNA technology, ecology, evolution, food deserts, and scientific inquiry.

PRESENTATIONS, SYMPOSIA & PROFESSIONAL ENGAGEMENTS

- Summer BIOME 2026 Participant and Presenter, Virtual | June 23-25, 2026. Presentation: “Enhancing Ecology: Digital Field Notes from a Changing Planet with AI and Science Gateways.” Focused on AI, Science Gateways, HPC, CUREs, and Ecology course redesign impacting approximately 45 biology majors annually.
- CDS-Exchange Symposium / Lightning Talk, Clark Atlanta University, Atlanta, GA | April 22-25, 2026. Shared curriculum goals integrating Jupyter Notebook, cloud-based modeling, data visualization, ethical AI use, and HBCU STEM pathway development.
- Presenter, HBCU AICon, Austin, TX | March 2026. Topic: “Strategies for AI Integration in HBCU STEM Curricula.”
- Presenter, AAC&U Annual Meeting, Washington, D.C. | January 2026. Topic: “Advancing Student Success through AI-Enhanced Instructional Models.”
- TSU AI Summit, Train-the-Trainer | 2026.
- Attendee, 2026 AI Impact Summit, Nashville, TN.
- Attendee, NSEA Symposium | 2026.
- Attendee, NAIRR CloudBank Webinar | 2026.
- Participant, NAIRR AI Education Webinar Series | 2026. Focus: classroom allocations, computing access, AI resources, and faculty/student opportunities for HBCUs.
- Future Professional Engagements | 2026-2027. Additional events to be added as titles, dates, and participation status are confirmed.

EDUCATION

- Ph.D., Urban Forestry, Environment & Natural Resources — Southern University and A&M College, 2021. Dissertation: Phytoremediation Efficacy of Eastern Cottonwood (Populus deltoides) to Heavy Metal Contamination.
- M.S., Biological Sciences — Southern University and A&M College, 2004. Thesis: Phytoremediation Potential and Response of Phaseolus vulgaris to Cadmium.
- A.S., Crime Scene Investigations/Forensics — Everest University (Online), 2008.
- B.S., Biological Sciences — Southern University and A&M College, 2001.

PROFESSIONAL CERTIFICATIONS & TECHNICAL TRAINING

- Google AI Certification (2026).
- Tennessee State University AI SuperUser Certification (2025).
- UC San Diego Faculty-Hack-a-Thon / SGX3 Advanced Computing Participant (2025).
- IBM Advanced AI (2024); IBM AI (2021); IBM Cybersecurity (2023); IBM IoT (2022); IBM Data Science & Design Thinking (2020).
- AWS Cloud Computing (2022); Apple Teacher — iPad & Mac (2018).
- Quality Matters — Applying the QM Rubric (2010); Improving Your Online Course (2020).

- Moodle (2016), Blackboard (2014), Online LMS Training, Online Course Development (2009-2010), TopHat Online Engagement Training (2018).

TECHNICAL PROFICIENCY & AI TOOLS

Canvas LMS, Moodle, NotebookLM, Google AI Studio, ChatGPT/Custom GPT Development, Microsoft Copilot, Perplexity, Napkin AI, Jupyter Notebook, Google Colab, learning analytics dashboards, rubric design, prompt engineering, responsible AI policy development, bias/accuracy evaluation, AI disclosure frameworks, mobile/iPad integration, accessible course design, and online/hybrid learning architecture.

MENTORING & ADVISING

- Faculty/Student Mentor & Advisor (2008-Present): academic planning, pre-health pathways, research placements, internships, graduate/professional school preparation, and conference readiness.
- Research Mentor and Thesis/Capstone Advisor (2025-2026): supports undergraduate honors theses and graduate capstone work integrating scientific inquiry, digital tools, literature synthesis, and data-enabled research methodologies.
- Pre-Med/Pre-Health Advisor; Faculty Advisor/Sponsor: MAPS, Sigma Xi, Beta Kappa Chi, Tri-Beta Biological Honor Society, and JagBites Dental Club.
- Upward Bound TRIO (2006-Present): environmental and green-tech projects including phytoremediation, soil/water toxicity, and STEM research mentorship.

SERVICE & COMMITTEES

- Institute of Teaching and Learning Excellence (ITLE) Committee; First 36 Pilot Program; Tablet PC Pilot Program.
- Planning/Steering Contributor: Beta Kappa Chi / National Institute of Science Annual Conference.
- Science Fair Judge (regional/state); Med School Admit (MSA) Program committee member and organizer.
- Faculty workshop contributor in RSI chart development, AI integration, accessible course design, and future-forward pedagogy.

RESEARCH & SCHOLARLY INTERESTS

Applied phytoremediation of contaminated urban/industrial soils; physiological and anatomical plant responses; environmental monitoring; biology education; AI-integrated STEM pedagogy; HBCU student success; computational/data literacy in biology; translational connections among environment, health, community, and STEMM learning.

PUBLICATIONS & ABSTRACTS (SELECTED)

- Thota, K.C., Kandel, S., Abdulkadir, A., Beals, M., D'Auvergne, O., Rosby, R., & Hossain, E. (2025). Phytochemical profiling and antimicrobial efficacy of Loropetalum chinense var. rubrum leaf crude extracts. (Abstract).
- Hossain, E., Tasnim, H., Rogers, B.T., Rosby, R., Beals, M., & D'Auvergne, O. (2023). NADPH oxidase inhibitor diphenyleneiodonium prevents arsenic-induced downregulation of ABCG1 in mouse aortic endothelial cells. International Journal of Toxicology, 42(1), 73-73. (Abstract).
- Beals, M.K., Miller, J.E., Olcott, D.D., LeBleu, M., Garber, J., & Foti, J. (2005). Integrated approach for controlling nematode parasites in small ruminants. BKX/NIS Joint National Meeting. (Abstract).
- Beals, M.K., & Tate, T.M. (2003-2004). Phytoremediation potential of Phaseolus vulgaris. BKX/NIS and SOT Southeastern Section. (Abstracts/Posters).

HONORS & MEMBERSHIPS

- Awards: 2nd Place (Oral), BKX/NIS Joint National Meeting (2003); 3rd Place (Oral), BKX/NIS (2002).
- Honor Societies/Organizations: Beta Kappa Chi; Sigma Xi; National Institute of Science; Beta Beta Beta; Endocrine Society; South Central Region SOT; MAPS; Alpha Kappa Alpha Sorority, Inc.

REFERENCES

Available upon request.